CSW + Cisco Secure Firewall Integration Guide

NetFlow (NSEL) visibility and FMC policy enforcement — customer reference

Cisco Secure Workload User Education

2026-06-04

Table of Contents

Cisco Secure Workload + Cisco Secure Firewall

NetFlow (NSEL) ingestion and policy enforcement on the firewall

> **Disclaimer:** Companion learning material — not official Cisco product documentation. Validate design, supported versions, and limits against your tenant, release notes, and:

- FMC Integration Guide (Cisco.com)
- Secure Workload & Firewall deep dive (secure.cisco.com)

Architecture diagrams: `docs/ARCHITECTURE.md` — customer-shareable Mermaid diagrams.

Executive summary

In hybrid multi-cloud environments, policy controls often fragment across host firewalls, network firewalls, SDN, and cloud security groups — managed by different teams. **Cisco Secure Workload (CSW)** and **Cisco Secure Firewall** together provide a unified microsegmentation path for workloads where **host agents cannot be installed**, while still supporting mixed agent + agentless estates.

Three integrated value streams:

Use case	What it does	Primary path
Microsegmentation	Discover agentless flows (NSEL), model policy (ADM), enforce on FTD via FMC	Secure Firewall Connector + FMC Connector
Virtual Patch	Export workload CVEs to FMC; apply compensating IPS rules	FMC Connector (Virtual Patch tab) + agents

Use case	What it does	Primary path
	until patches ship	
Rapid Threat Containment	FMC detects anomaly → Remediation Module → CSW quarantine guardrails	FMC Remediation Module + CSW labels

Solution overview

CSW delivers three platform capabilities that intersect with Secure Firewall:

1. **Zero Trust Microsegmentation** — agent and agentless discovery, ADM-based policy, enforcement on host firewalls, DPUs, **network firewalls**, load balancers, and cloud controls.
2. **Vulnerability Detection and Protection** — runtime CVE visibility from agents; **virtual patch via Secure Firewall IPS**.
3. **Behavioral Detection and Protection** — process/forensic analytics; **Rapid Threat Containment** with FMC correlation + CSW guardrails.

Architecture (customer reference)

High-level — two paths, one platform

```
flowchart TB
  subgraph Visibility["Visibility — NSEL"]
    FTD["Secure Firewall FTD/ASA"]
    SFC["Secure Firewall Connector<br/>Ingest Appliance · UDP 4729"]
    FTD -->|"NSEL NetFlow v9"| SFC
  end

  subgraph CSW["Cisco Secure Workload"]
    ADM["Flow Analysis · ADM · Labels"]
    PE["Policy Engine"]
    SFC --> ADM --> PE
  end

  subgraph Enforcement["Enforcement — FMC"]
    FMCc["FMC Connector · HTTPS 443"]
    FMC["FMC / cdFMC"]
    PE -->|"Dynamic Objects + ACP rules"| FMCc --> FMC
    FMC -->|"auto-deploy"| FTD2["FTD devices"]
  end

  CMDB["CMDB · IPAM · Manual labels"] --> ADM
```

SaaS CSW: Secure Connector tunnels traffic between CSW cloud, Ingest appliance, and on-prem FMC.

See `docs/ARCHITECTURE.md` for sequence diagrams, scope mapping, insertion options, virtual patch, and threat containment flows.

Two connectors — do not mix them

Integration	Connector	Purpose	Data direction
Flow visibility (NetFlow / NSEL)	**Secure Firewall Connector** on **Ingest appliance**	CSW sees flows from FTD/ASA without agents	Firewall → Ingest → CSW
Policy enforcement	**FMC Connector** (Cisco Secure Firewall in UI)	CSW pushes segmentation to FTD via FMC	CSW → FMC → FTD

Also available	Use when
NetFlow connector	Switches/routers export NetFlow v9/IPFIX — **not** FTD NSEL
Host agents	Process-level visibility + host enforcement
ACI connector	Fabric policy — separate from this FTD/FMC guide

Workload protection level (design first)

Before choosing firewall insertion mode, define the **security/trust boundary** your organization uses (subnet, zone, app tier, persona). The white paper recommends aligning network engineers and app owners on a measurable protection level — commonly **subnet-level** for agentless FTD insertion.

Persona	Typical boundary	Agent vs agentless
App / DevOps	App tier, namespace	Often agent-based
Network / Firewall	Subnet, zone, VRF	Often agentless (this guide)
Cloud	VPC/VNet, security group	Mixed

This drives whether you use **L2 transparent** (fine-grained), **L3 routed** (faster zone segmentation), or **cloud hub** patterns — see Insertion options.

Prerequisites checklist

CSW + Ingest (visibility)

#	Requirement	Notes
1	CSW tenant (SaaS or on-prem) with admin access	Scopes and workspaces ready
2	**Secure Workload Ingest** virtual appliance	**Active** before enabling connector
3	**Secure Connector** (SaaS) or network path	Mandatory for SaaS
4	FTD/ASA can reach Ingest on **UDP 4729**	Copy IP:port from connector details
5	Capacity	**45k fps** per connector; **135k fps** per Ingest appliance (on-prem)

FMC (enforcement)

#	Requirement	Notes
1	**FMC** or **cdFMC** with supported FTD devices	FTD managed by FMC only
2	**FMC REST API user** with **administrative privileges**	Required for connector
3	FTD associated with FMC; ACP assigned; deploy verified	Test FMC→FTD deploy before CSW
4	**Secure Connector** if SaaS CSW cannot reach FMC directly	HTTPS **443** to FMC
5	FMC HA: include **standby FMC IP** in connector	Auto-switch on failover
6	Supported versions	Segmentation: FMC **7.0.x+** (dynamic objects **7.0.1+**); Virtual Patch: **7.2+**

Virtual Patch (optional)

- CSW **agents** on target workloads
- Threat intelligence **CVE data pack** on CSW cluster
- Workloads use **FTD as default gateway** (traffic through IPS path)

- FMC network discovery sees hosts under **Analysis → Network Map**

Part A — NetFlow / NSEL visibility (Secure Firewall Connector)

Goal: CSW receives bidirectional, stateful flow records from the firewall **without** host agents.

Step A1 — Plan ingest placement

1. **Manage → Workloads → Virtual Appliances** — confirm Ingest is **Active**.
2. Record **connector slot IP** and port (**4729** default).
3. Open firewall path: **UDP 4729** from FTD/ASA to Ingest.
4. **VRF: Manage → Agents → Configuration → Agent Remote VRF Configurations** — one connector per VRF.

Videos: Connector Overview · Connector Deployment

Step A2 — Enable Secure Firewall Connector in CSW

1. **Manage → Workloads → Connectors**.
2. Select **Cisco Secure Firewall Connector** (formerly ASA connector).
3. **Enable** on **Ingest appliance** (not Edge-only).
4. Record connector **ID** and **IP:port** from connector details page.

Limits: 1 Secure Firewall connector per Ingest appliance; **10** per tenant.

Step A3 — Enable NSEL on Secure Firewall

ASA example (replace <INGEST_CONNECTOR_IP>):

```
flow-export destination outside <INGEST_CONNECTOR_IP> 4729
flow-export template timeout-rate 1
!
policy-map flow_export_policy
  class class-default
  flow-export event-type flow-create destination <INGEST_CONNECTOR_IP>
  flow-export event-type flow-teardown destination
<INGEST_CONNECTOR_IP>
  flow-export event-type flow-denied destination <INGEST_CONNECTOR_IP>
  flow-export event-type flow-update destination <INGEST_CONNECTOR_IP>
  user-statistics accounting
service-policy flow_export_policy global
```

FTD: Configure NSEL export via FMC/device template — ASA NetFlow Implementation Guide.

Videos: Part 1 · Part 2 · Part 3 · **2025–2026 updates**

Step A4 — Validate flow ingestion

1. Generate permitted and **denied** test traffic.

2. **Flow Analysis** — confirm firewall-sourced endpoints.
3. Connector heartbeat healthy under **Manage** → **Workloads** → **Connectors**.
4. Optional API filters: `user_src_NETFLOW_IDENTIFIED` / `user_dst_NETFLOW_IDENTIFIED`.

NSEL handling: flow-create/update → exported; flow-denied → **rejected** disposition; connector sends **forward + reverse** flows; **NAT flow-stitching** for end-to-end visibility.

Video: Flow Analysis

Step A5 — Label workloads and run discovery

1. Import **labels** from CMDB (ServiceNow), IPAM, or manual assignment.
2. Run **ADM** on scoped workspace.
3. Remain in **Monitor** mode until app owners validate dependencies.

Videos: Basic Application Discovery · Enhancing Application Discovery · ADM & Policy Analysis

---

Part B — Policy enforcement on Secure Firewall (FMC Connector)

Goal: CSW pushes **L3/L4 segmentation** to FTD via FMC using **Dynamic Objects —** inventory changes refresh membership without manual object edits.

> CSW orchestrated rules are **L3/L4 east-west microsegmentation**. Layer-7 FMC features should not be added to CSW-controlled rules (not supported).

Step B1 — Onboard firewalls in FMC

1. FTD devices registered to **FMC** or **cdFMC**.
2. Each FTD assigned to an **Access Control Policy (ACP)**.
3. Verify FMC can **deploy** to FTD and traffic behaves as expected.
4. Document **one scope** → **one ACP** mapping plan (1:1 only).

Video: FMC Integration (Edge / Ingest / Appliance)

Step B2 — Create FMC Connector in CSW

1. **Manage** → **Workloads** → **Connectors** → **Firewall** → **Cisco Secure Firewall**.
2. Click **Configure your new connector here**.
3. Complete **New Connection** fields:

Field	Description
Connector Name	Unique name for this FMC connection
Username / Password	FMC REST API credentials (admin privileges)
CA Certificate	FMC CA cert for TLS validation, or **Disable SSL** on trusted networks

Field	Description
Server IP/FQDN and Port	FMC hostname or IP (**443**)
HTTP Proxy	If required: `<proxy.host>:<proxy.port>`
Secure Connector	Enable when SaaS CSW reaches FMC through Secure Connector

4. Click **Create**; verify **Event Log** tab for errors.

Official procedure: FMC Integration Guide — Connection Settings

Step B3 — Map scope to Access Control Policy

1. Open FMC connector → **Segmentation** tab → **+ Add**.
2. **Add ACP Mapping:**
    - Select **Access Policy** (ACP) from FMC.
    - Map to exactly **one Scope**.
    - **Use Secure Workload Catch-All** — CSW catch-all in Default section, or use FMC default action.
    - **Enforcement Mode:**
        - **Merge** — CSW rules coexist with FMC rules; configure priority.
        - **Override** — CSW replaces user-created rules in mapped sections.
    - **Absolute / Default Policies priority** (Merge only):
        - Insert **above** or **below** existing Mandatory/Default rules per section.
3. Click **Submit**.

Scope mapping strategy

Model	When	Behavior
Child / leaf scope	Single application behind firewall	Push that app's policies + parent guardrails
Parent scope	Multiple applications	Push parent + eligible child policies (hierarchical)

Topology awareness

CSW pushes rules only to FTD devices on the **traffic path** for the mapped scope — avoids rule sprawl across unrelated firewalls.

Videos: Policy Enforcement Overview · Where to Enforce · Policy Ordering

Step B4 — What CSW creates in FMC

When enforcement is enabled on a workspace, CSW converts policies to FMC rules:

FMC artifact	Prefix	ACP section	Purpose
Golden rules	`Workload_golden__`	Mandatory	CSW ↔ agent communication

FMC artifact	Prefix	ACP section	Purpose
			behind FW
Segmentation rules	`Workload__`	Mandatory / Default	Enforced workspace policies
Catch-all rules	`Workload_ca__`	Default	If **Use Secure Workload Catch-All** enabled
Dynamic objects	`WorkloadObj__`	Objects	Scope/inventory IP membership

CSW policy type	FMC placement
Absolute	Mandatory category
Default	Default category

> **Do not manually edit** `Workload_*` rules in FMC — CSW restores them on next push. Independent FMC rules are safe in **Merge** mode if they avoid these prefixes.

Step B5 — Model policy, then enforce

1. Build or accept **ADM-recommended** policies in workspace tied to scope.
2. **Simulate / Monitor** — Policy Visual and flow analysis.
3. Change window: **enable enforcement** on workspace.
4. Confirm FMC deployment succeeds; verify Dynamic Objects update as inventory changes.

Videos: Policy Lifecycle · Policy Validation and Analysis

Step B6 — Validate enforcement

Test	Expected result
Allowed business flow	Passes; CSW/FMC show permit
Intentionally denied flow	Blocked at FTD; CSW rejected-flow evidence
New workload in scope	`WorkloadObj__` membership updates
FMC deployment	Automatic on CSW policy change
Compliance monitoring	Alerts on rejected flows / policy deviation

Check **Event Log** on FMC connector for push errors (Information / Warning / Error).

Part C — Virtual Patch (optional)

Compensating **IPS** control for CVEs until permanent patches are deployed.

1. FMC connector → **Virtual Patch** tab → **Create a Virtual Patching Rule**.
2. Define **Host Query** (scope/filter) and **CVE Query** (CVSS, Cisco Risk Score, etc.).
3. CSW exports matching CVEs to FMC **Third-Party Vulnerabilities**.
4. FMC admin runs **Cisco Recommended Rules** to map CVEs → Snort signatures.
5. Apply IPS policy on ACP rule for affected traffic flows.

Prerequisites: Agents on workloads; CVE data pack; FTD as gateway; FMC **7.2+**.

Part D — Rapid Threat Containment (optional)

Automated quarantine when FMC detects malicious behavior:

1. Install **FMC Remediation Module for Secure Workload** on FMC.
2. Configure CSW cluster IP, root scope, API key (**User Data Upload** permission).
3. Define **Correlation Rules** and **Correlation Policy** responses.
4. On trigger, Remediation Module sends affected IPs to CSW via API.
5. CSW **guardrail policies** (quarantine labels) enforce across agents and FTD.

Firewall insertion options

Mode	Segmentation	NSEL visibility	Best fit
L2 Transparent	Intra- + inter-subnet	Full	Legacy OS, fine-grained, localized
L3 Routed	Inter-subnet only	Inter-subnet only	Quick zone segment, dev/non-prod
ACI Service Graph PBR	Intra + inter EPG/ESG	Full (FW in path)	ACI + FTD multi-policy
AWS hub VPC	Inter-VPC	VPC logs + NSEL	Centralized cloud east-west
AWS distributed	Intra-VPC	VPC logs + NSEL	Per-VPC enforcement
Azure hub VNet	Intra- + inter-VNet	NSG logs + NSEL	UDR steering
GCP hub VPC	Inter-VPC	VPC logs + NSEL	Centralized GCP

All patterns support **dual management**: CSW east-west + FMC north-south on the same ACP (Merge mode).

FAQs (from Cisco white paper)

Question	Answer
Mixed agent + agentless behind same mapped firewall?	Works. Agent-based rules may **also** appear in ACP (hierarchical policy). Primary use case remains agentless protection.
Map multiple scopes to one ACP?	**No** — strictly **1 scope : 1 ACP**.
Add Layer-7 / FMC features to CSW rules?	**Not supported** — CSW rules are L3/L4 east-west only.
Dual-management (CSW + FMC rules)?	**Merge** mode + rule ordering; preserve order after go-live.
Modify a CSW-owned rule in FMC?	CSW **overwrites** on next push.
Supported FMC versions?	Dynamic objects: **7.0.1+**; Virtual Patch: **7.2+**; qualified: **7.0.1**, **7.2.5**.

Recommended video learning path

Order	Video	Link
1	Connector Overview	Watch
2	Connector Deployment	Watch
3	Firewall Integration Part 1	Watch
4	Part 2	Watch
5	Part 3	Watch
6	**2025–2026 integration updates**	Watch
7	FMC + Edge / Ingest / Appliance	Watch
8	Where to Enforce	Watch
9	Policy Enforcement Overview	Watch

NetFlow connector vs Secure Firewall connector vs API flowsearch

Method	Use when	Scale notes
Secure Firewall Connector (NSEL)	FTD/ASA agentless visibility	**45k fps** per connector; **135k fps** per Ingest
NetFlow connector	Switches/routers	~15k fps per connector

Method	Use when	Scale notes
	(v9/IPFIX)	
Host agents	Process-level + host enforce	Best fidelity
OpenAPI `flowsearch`	Query ingested flows	Paginate; ≤24h; 1h chunks — not bulk ingest

POV evidence checklist

Evidence	Proves
Connector enabled + heartbeat healthy	Ingest path live
Sample flows in Flow Analysis (NSEL)	Visibility
ADM map for scoped app	Discovery value
FMC connector connected + Event Log clean	Enforcement path
Scope ↔ ACP mapping screenshot	Topology binding
`Workload__` / `WorkloadObj__` in FMC	Policy push working
Monitor-mode policy stats	Safe pre-enforce review
Post-enforce deny test + FMC hit	Enforcement works
Allowed transaction test	No business break

Common pitfalls

Pitfall	Fix
Confusing **NetFlow connector** with **Secure Firewall connector**	NSEL from FTD/ASA → Secure Firewall Connector only
Expecting **NSEL alone** to enforce	Requires **FMC connector** + workspace enforce
Editing `Workload_*` rules in FMC	CSW will overwrite — change policy in CSW
Multiple scopes on one ACP	Redesign — **1:1 mapping only**
L3 routed FW for intra-subnet segment	Use **L2 transparent** or host agents
Bulk **`flowsearch`** API timeouts	Paginate; chunk time windows
Skipping **Secure Connector** on SaaS	Required for Ingest and FMC reachability

Official Cisco documentation

Document	URL
FMC Integration Guide	cisco-secure-workload-and-fmc-integration-guide.html
Deep dive white paper	secure-workload-whitepaper
CSW 4.0 SaaS — Connectors	m-connectors.html
Marketing white paper	sec-workload-firewall-wp.html
ASA NSEL configuration	ASA NetFlow Implementation Guide

Related repos

- CSW-Secure-Firewall-Integration-Guide — **this repo**
- CSW-User-Education — video library and onboarding
- CSW-Policy-Lifecycle — ADM → Monitor → Enforce
- CSW-Agent-Installation-Guide — host agents
- CSW-Identity-Integration-Guide — AD / Entra identity
- CSW-ServiceNow-Connector-Guide — CMDB labels
- csw-splunk-integration — SIEM
- CSW-Compliance-Mapping — compliance frameworks
- csw-logs-check — host enforce timing from agent logs

CSW + Secure Firewall — Architecture Reference

Customer-facing architecture diagrams and design reference, derived from the Cisco Secure Workload & Secure Firewall deep dive and the FMC Integration Guide.

> **Disclaimer:** Companion material — validate against your CSW and FMC release notes before production.

---

1. High-level integration (microsegmentation)

This is the reference architecture for **agentless east-west segmentation** using NSEL visibility and FMC enforcement.

```
flowchart TB
  subgraph Sources["Context & telemetry"]
    CMDB["CMDB / ServiceNow"]
    IPAM["IPAM / DNS"]
    Labels["Manual labels"]
  end

  subgraph Workloads["Application workloads"]
```

```
    Agent["Agent-based workloads<br/>host firewall enforce"]
    Agentless["Agentless workloads<br/>legacy / appliance / no agent"]
  end

  subgraph Visibility["Visibility path — NSEL"]
    FTD["Secure Firewall FTD / ASA<br/>Transparent or Routed mode"]
    NSEL["NSEL export<br/>NetFlow v9 · UDP 4729"]
    Ingest["Secure Workload Ingest Appliance"]
    SFC["Secure Firewall Connector<br/>Docker · up to 45k fps each"]
    FTD -->|"flow-create · teardown · deny · update"| NSEL --> SFC
    SFC --> Ingest
  end

  subgraph CSW["Cisco Secure Workload"]
    FlowDB["Flow analysis & inventory"]
    ADM["Application Dependency Mapping"]
    Policy["Policy engine<br/>Monitor → Simulate → Enforce"]
    Ingest --> FlowDB
    CMDB --> FlowDB
    IPAM --> FlowDB
    Labels --> FlowDB
    FlowDB --> ADM --> Policy
  end

  subgraph Enforcement["Enforcement path — FMC"]
    FMCConn["FMC Connector<br/>HTTPS REST API :443"]
    FMC["Secure Firewall Management Center"]
    ACP["Access Control Policy<br/>Dynamic Objects"]
    Policy -->|"Scope ↔ ACP mapping<br/>Topology-aware rules"| FMCConn
    FMCConn --> FMC --> ACP
    ACP -->|"auto-deploy"| FTD
  end

  Agentless -.->|"traffic through FW"| FTD
  Agent -.->|"optional: rules also pushed to ACP"| Policy
```

Key idea: Visibility and enforcement use **different connectors**. NSEL tells CSW what flows exist; the FMC connector pushes **L3/L4 segmentation policy** to FTD when you enforce a workspace.

---

2. Data flow — visibility (NSEL)

```
sequenceDiagram
  participant W as Workload traffic
  participant FW as Secure Firewall FTD/ASA
  participant IC as Secure Firewall Connector
  participant CSW as Secure Workload
```

```
  W->>FW: Application flow
  FW->>FW: Stateful inspection
  FW->>IC: NSEL record (create/update/deny/teardown)
  Note over IC: Decapsulate NetFlow v9<br/>Forward + reverse flows<br/>NAT flow-stitching
  IC->>CSW: Flow observations (agentless)
  CSW->>CSW: ADM · labels · policy modeling
```

NSEL capability	Customer value
Stateful events	See permits and **denies** — not just sampled NetFlow
Bidirectional flows	Connector exports forward + reverse observations
Flow stitching	End-to-end visibility when **NAT** is in path
Agentless discovery	No agent on legacy OS, appliances, or restricted hosts

Scale (on-prem Ingest, per Cisco white paper):

Metric	Limit
Per Secure Firewall connector	Up to **45,000 fps**
Per Ingest appliance (total)	Up to **135,000 fps**
Connectors per Ingest appliance	**1** Secure Firewall connector
Connectors per tenant	**10**

3. Data flow — enforcement (FMC)

```
sequenceDiagram
  participant Op as Operator
  participant CSW as Secure Workload
  participant FMC as FMC / cdFMC
  participant FTD as FTD cluster

  Op->>CSW: Enable enforcement on workspace
  CSW->>CSW: Resolve scope inventory → IP sets
  CSW->>FMC: Create/update Dynamic Objects (WorkloadObj_*)
  CSW->>FMC: Push ACP rules (Workload_* · Workload_golden_* · Workload_ca_*)
  FMC->>FTD: Deploy policy (no manual redeploy for CSW changes)
  Note over FTD: Topology awareness — rules only on firewalls in traffic path
```

FMC rule prefixes (do not edit in FMC)

Prefix	Purpose	ACP section
`Workload_golden__`	Allow CSW ↔ agents behind the firewall	Mandatory
`Workload__`	Segmentation rules from enforced workspaces	Mandatory / Default
`Workload_ca__`	Catch-all rules (if **Use Secure Workload Catch-All** enabled)	Default
`WorkloadObj__`	Dynamic objects for scope/inventory membership	Objects

> CSW **overwrites** changes to rules using these prefixes on the next push. Independent FMC rules are preserved in **Merge** mode if they do not use these prefixes.

---

4. Scope ↔ ACP mapping models

```
flowchart LR
  subgraph Single["Single application"]
    CS1["Child / leaf scope"]
    ACP1["One ACP"]
    CS1 -->|"1:1 mapping"| ACP1
  end

  subgraph Multi["Multiple applications"]
    Parent["Parent scope"]
    C1["Child scope A"]
    C2["Child scope B"]
    ACP2["One ACP"]
    Parent --> C1
    Parent --> C2
    Parent -->|"hierarchical policy"| ACP2
  end
```

Mapping	When to use	What CSW pushes
Child scope → ACP	One application protected by this firewall pair	Policies for that app + parent guardrails
Parent scope → ACP	Multiple apps behind same firewall estate	Parent + eligible child scope policies

Constraint: One ACP maps to **one** scope only (1:1). Plan ACP boundaries accordingly.

---

5. ACP mapping options

Setting	Merge mode	Override mode
Existing FMC rules	Kept; CSW rules inserted top or bottom	Replaced by CSW rules
Rule priority	Configurable: above/below Mandatory and Default sections	N/A
Catch-all	Optional: CSW catch-all **or** FMC default action	Same
Dual management	Yes — network team keeps north-south / baseline rules	CSW-only in mapped sections

Policy type placement:

CSW policy type	FMC ACP section
Absolute policies	**Mandatory** category
Default policies	**Default** category

6. Firewall insertion options (on-premises)

Choose insertion based on **protection level** (typically subnet / zone boundary) and segmentation granularity.

```
flowchart TB
  subgraph L2["Layer 2 — Transparent / bump-in-wire"]
    L2W["Fine-grained · intra + inter-subnet<br/>Full NSEL
visibility<br/>Legacy OS / localized workloads"]
  end

  subgraph L3["Layer 3 — Routed / gateway"]
    L3W["Faster time-to-segment · inter-subnet only<br/>Partial NSEL
(no intra-subnet)<br/>Dev / non-prod zones"]
  end

  subgraph ACI["ACI + Service Graph"]
    ACIW["Intra + inter EPG/ESG<br/>PBR redirect or
go-to/through<br/>CSW + FMC + ACI policies"]
  end
```

Insertion	Segmentation	NSEL visibility	Best for
L2 Transparent	Intra- + inter-subnet	Full (intra + inter)	Fine-grained, legacy OS, localized apps

Insertion	Segmentation	NSEL visibility	Best for
L3 Routed	Inter-subnet only	Inter-subnet only	Quick zone segmentation, distributed workloads
ACI Service Graph (PBR)	Intra + inter EPG/ESG	Full with FW in path	ACI fabric + FTD service graph
ACI Go-To / Go-Through	Inter or intra+inter	Varies by mode	North-south or strict in-path designs

7. Cloud insertion options (summary)

Platform	Pattern	East-west protection	Visibility
AWS	Centralized hub VPC FTD	Inter-VPC / inter-subnet	VPC flow logs + NSEL
AWS	Distributed per-VPC FTD	Intra-VPC inter-subnet	VPC flow logs + NSEL
Azure	Hub VNet FTD + UDR	Intra- + inter-VNet	NSG flow logs + NSEL
GCP	Hub VPC FTD	Inter-VPC	VPC flow logs + NSEL

All cloud patterns support **FMC dual management**: CSW owns east-west microsegmentation; FMC owns north-south ingress/egress.

---

8. Extended use cases

Virtual Patch (agent + FMC)

```
flowchart LR
  Agent["CSW agent on workload"] -->|"CVE inventory"| CSW
  CSW -->|"FMC Connector Virtual Patch tab"| FMC
  FMC -->|"Third-Party Vulnerabilities + Snort mapping"| IPS["IPS policy / Cisco Recommended Rules"]
  IPS --> FTD
```

- Requires **agents** on workloads where virtual patch applies.
- Workloads must use an **FTD address as default gateway** for traffic to hit IPS.
- FMC **7.2+** for virtual patch; CSW publishes CVEs → FMC → fine-tuned IPS rules.

Rapid Threat Containment

```
flowchart LR
  Anomaly["Malware / IPS / correlation event"] --> FMC
  FMC -->|"Remediation Module API"| CSW
  CSW -->|"Quarantine label / guardrail policy"| Enforce["Host agent +
FTD enforce"]
```

- FMC **Remediation Module** sends affected IPs to CSW.
- CSW **guardrail policies** (label-based) quarantine workloads across enforcement points.

---

9. Connectivity requirements

Path	Protocol	Port	Notes
FTD/ASA → Ingest	UDP (NSEL)	**4729** (default)	Firewall must reach Ingest slot IP
CSW → FMC	HTTPS	**443**	REST API; admin credentials
SaaS CSW ↔ on-prem	Secure Connector	tunneled	Required for SaaS + on-prem FMC/Ingest
FMC HA	Include **standby FMC** IP	443	Auto-failover to new active FMC

10. Supported versions (reference)

Capability	CSW	FMC / FTD
Segmentation (ACP mapping, topology)	3.9.1.1+	**7.0.x+** (dynamic objects: **7.0.1+**)
Virtual Patch	3.8.1.1+	**7.2+**
Simplified segmentation workflow	3.9.1.1+	7.2 recommended
Engineering qualified (white paper)	—	**7.0.1**, **7.2.5**

Platforms: FTD devices **managed by FMC only** (not standalone ASA for enforcement path). FTD **Transparent** and **Routed** modes supported.

Official references

Document	URL
Deep dive white paper	secure.cisco.com/secure-workload/

Document	URL
	docs/secure-workload-whitepaper
FMC integration guide	cisco-secure-workload-and-fmc-integration-guide.html
Marketing white paper	sec-workload-firewall-wp.html